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REVIEW PAPER

Microfluidics-driven high-throughput phenotyping and screening in synthetic biology: from single cells to cell-free systems

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Abstract
The interdisciplinary nature of synthetic biology merges engineering principles with biology and provides innovative solutions for issues in the biomanufacturing industry. To develop industrially applicable biocatalysts and/or microbial cell factories, a design-build-test-learn cycle-based iterative process is necessary, which is often time-consuming and labor-intensive. The integration of microfluidic technologies into synthetic biology can accelerate these processes, particularly for achieving high-throughput phenotyping and screening. In this review, we examine the potential of microfluidic technologies to revolutionize synthetic biology. Although commercial microfluidics demonstrate superior throughput for single-cell assays, their application can be limited, for example, in cases where products are retained inside the cells. Droplet microfluidics, on the other hand, is a rather flexible platform and shows high diversity in single-cell, cell-to-cell interaction-based, and cell-free reaction-based analyses. By examining previous studies, we have summarized the potential of microfluidic technologies to foster sustainable biomanufacturing and advanced biological engineering.

Keywords Microfluidics · Synthetic biology · High-throughput screening · Biofoundry

1 Introduction

Synthetic biology is an interdisciplinary field that integrates engineering and computational principles with fundamental components of biology. It plays a pivotal role in bio-manufacturing, acting as a critical link to the bioeconomy and offering promising solutions to global challenges [1, 2]. The primary goal of synthetic biology is to design and construct new biological systems or to modify existing systems to introduce novel functionalities and applications. This paradigm shift has replaced traditional biotechnology practices with a more reliable and scalable industry that facilitates carbon–neutral production with diverse and sustainable outputs [3]. Research in this domain often focuses on developing and engineering biocatalysts and cell factories through iterative cycles to design and construct large-scale genetic variable libraries, followed by phenotyping and screening.

Biofoundries play a crucial role by incorporating advanced automation technologies and computer-aided design and analysis, effectively accelerating the design-build-test-learn (DBTL) cycle while minimizing human errors and biases [4, 5]. Consequently, in addition to the fast and accurate construction of extensive genetic variations, high-throughput analysis and the selection of targets of interest, such as enzymes with high activity or cell factory strains that produce desired products at high titers, have become increasingly important. Moreover, the development of artificial intelligence in synthetic biology enables data-driven approaches, necessitating the large-scale acquisition of sequence-function data to efficiently explore and understand target biosystems [6]. This emphasizes the need for rapid phenotyping and library screening to advance sustainable biomanufacturing.

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One of the primary challenges is our limited ability to predict the phenotypes of modified biological systems. Current approaches predominantly rely on fluorescence-based detection methods that utilize fluorescent proteins and transcription factor- or riboswitch-based sensors [7]. Fluorescence-activated cell sorting (FACS) is a widely used commercial microfluidic device that offers unparalleled throughput and sensitivity for the analysis and sorting of single cells [8]. Another confinement-based commercial microfluidic system called the Beacon® Optofluidic System, while offering lower throughput, enables tracking of the phenotype and genotype of single cells during cultivation [9]. Additionally, droplet microfluidics, a promising approach that encapsulates individual samples within droplets, provides better throughput and potential for high-throughput phenotyping and screening applications [10, 11]. Nevertheless, further advancements are necessary to enhance the speed and accuracy of analysis and sorting to fully maximize the potential of droplet microfluidics for such applications.

Cell-free protein synthesis (CFPS) involves in vitro transcription and translation reactions, providing an efficient and rational approach for protein engineering and enzyme evolution that bypasses the complexities of cellular systems [12, 13]. CFPS has the potential to accelerate DBTL cycles by enabling rapid prototyping without the need for gene transfer processes [14]. The integration of droplet microfluidics with CFPS offers a valuable platform for enhancing the efficiency and effectiveness of the DBTL cycle as it allows precise control over reaction volumes and increased throughput with minimal reagent consumption, thus facilitating new discoveries and maximizing the potential of synthetic biology [15, 16].

Significant progress has been made in the development of alternative fluorescence- and label-free detection methods. Although these methods generally exhibit slightly lower throughput than fluorescence-based methods, they offer broader applicability by eliminating the need for limited labeling molecules or specific target biosensors. Recent advancements in Raman spectroscopy-based methods have demonstrated high accuracy and processing speed for label-free detection, making them valuable for high-throughput phenotyping and screening applications [17, 18].

In this review, we provide a comprehensive overview of the advancements in the potential of microfluidic technologies for high-throughput phenotyping and screening in synthetic biology (Fig. 1). First, we introduce rapid single-cell analysis using commercial microfluidic instruments and droplet microfluidics. Next, we discuss the potential of integrating droplet-based microfluidics with cell-free systems to enhance the efficiency of DBTL in biofoundries. In addition, we examine the development of label-free analytical methods. Overall, this review highlights the significant role of these technologies in advancing the field of engineering biology toward sustainability.

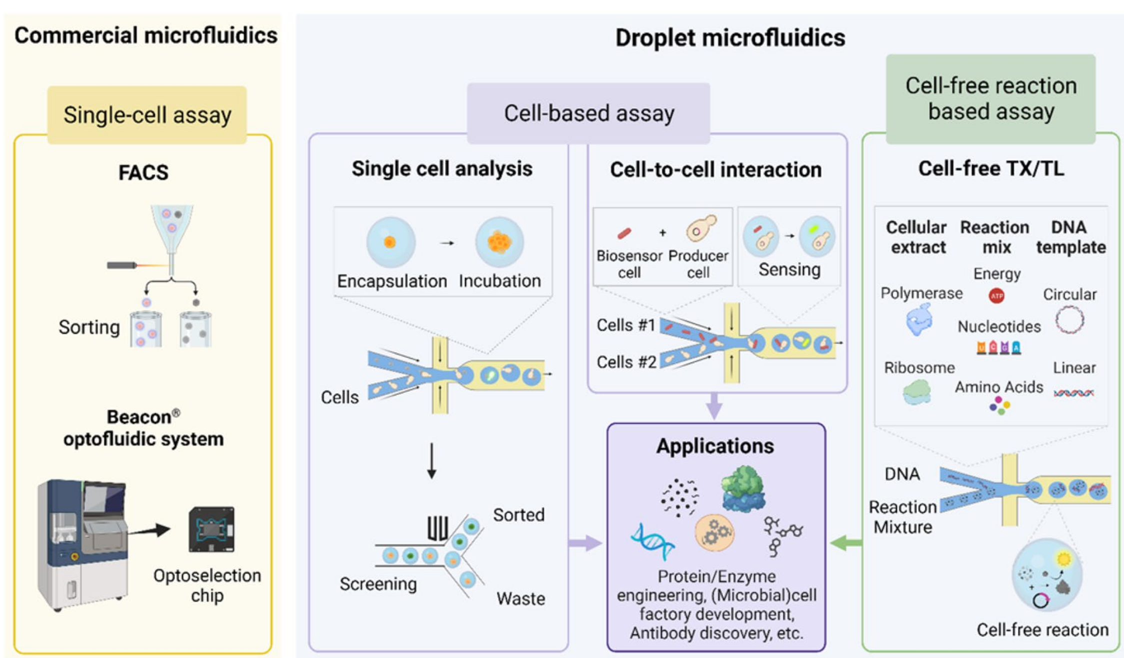

Fig. 1 Commercial and droplet microfluidics for high-throughput phenotyping and screening in synthetic biology applications. *FACS* fluorescence-activated cell sorting

2 Commercial microfluidic instruments for rapid single-cell prototyping and screening

FACS is an advanced and highly efficient microfluidic device used for the rapid analysis and sorting of cells based on their size and fluorescence properties using lasers. This technology has an impressive processing speed of up to 10^5 cells/s, making it the fastest available method for single-cell analysis [8]. Single-cell analysis leverages the inherent compartmentalization of individual cells, with the cell membrane acting as a protective barrier that traps fluorescent reporter molecules within a small cellular volume, enabling sensitive and reliable analysis. Consequently, single-cell analysis assisted by FACS has gained widespread use in the high-throughput screening of extensive libraries targeting the engineering of proteins, enzymes, and microbial strains over extended periods [19, 20].

To facilitate FACS-based single-cell analysis, genetic biosensors based on transcription factors [21–26] and riboswitches [27–31] have been utilized to measure the target products from biosynthesis pathways or enzymatic reactions. For example, the Genetic Enzyme Screening System (GESS) employs the transcriptional activator DmpR that is responsive to phenolic compounds. This system enables the screening of diverse enzymes, including tyrosine phenol-lyase, lipase, cellulase, and methyl parathion hydrolase [24, 32]. When combined with FACS, GESS effectively identified novel phosphatase genes, demonstrating its efficacy. Moreover, the recent integration of machine learning techniques has further improved the specificity and performance of the GESS [33].

The Beacon® Optofluidic System, developed by Berkeley Lights, Inc. (BLI), is a remarkable commercial microfluidic device designed specifically for confinement-based screening [9, 34]. Automated light-actuated dielectrophoresis (DEP) allows the precise positioning of individual cells in specific target locations, enabling the simultaneous cultivation of single cells in separate compartments with easy and precise manipulation of liquid samples, including nutrients. Beacon enables the study of individual cells with identical or distinct genotypes through growth monitoring and fluorescence-based imaging assays. Importantly, it streamlines the screening workflow by eliminating labor-intensive and time-consuming sorting and isolation processes [35].

Beacon is extensively used in rapid screening of diverse mammalian cell lines for antibody discovery [36] and biologics production [37]. Furthermore, Beacon shows promise for application in synthetic biology as a high-throughput screening system for engineering microbial cell factories that secrete small molecules [38]. Notably, Amyris and BLI collaborated to develop *Saccharomyces cerevisiae* to improve small molecule production [35].

Despite their superior throughputs and numerous successful applications, commercial devices have certain limitations. FACS demonstrates superior throughput; however, its application is limited because it cannot be used to track temporal processes in single cells and/or study processes that extend beyond the boundaries of individual cells, such as the secretion of products or cell-to-cell interactions. Beacon overcomes such drawbacks, but its throughput is relatively low, and it takes longer to analyze large-scale libraries (although Beacon shows better throughput than conventional microplate-based analysis). Alternatively, droplet microfluidics offers a promising approach for addressing the disadvantages of commercial devices. The comparison between commercial and droplet microfluidics, considering their advantages and disadvantages in synthetic biology applications, is summarized in Table 1.

3 Droplet microfluidics and cell-based assays

Droplet-based microfluidics have garnered significant attention from the Global Biofoundry Alliance. For instance, the Agile Biofoundry emphasizes the "microfluidics functionality" which expands its functionality for rapid gene synthesis, assembly, and high-throughput testing [11]. Meanwhile, the DAMP Lab (Design, Automation, Manufacturing, and Processes Laboratory) focuses on enhancing "microfluidics accessibility" through standardization to ensure high reproducibility of devices. Predictable fluid behavior at the microscale level will help in the efficient design of new signaling platforms. Their efforts are directed toward designing and fabricating user-friendly and cost-effective devices that can be operated by individuals without specialized expertise [39–41]. These advancements underscore the ongoing expansion and diversification of droplet-based microfluidics and establish it as a promising technology with broad potential across various research domains and biofoundries.

Exploiting water–oil immiscibility, droplet-based microfluidics generates uniform droplets within microchips, enabling rapid parallel processing and versatile applications beyond current limitations. These applications include screening cell-variant libraries for secreted proteins and enzymes, production of toxic chemicals, and engineering of enzymes for impermeable substrates [42–50]. For example, Huang et al. [42] conducted high-throughput α-amylase secretion screening by encapsulating UV-generated mutant libraries in droplets containing single cells with a fluorogenic substrate. Similarly, Pourmasoumi et al. [43] employed droplet-based fluidics to screen megasynthetase mutants,

Table 1 Comparison of commercial and droplet microfluidics in synthetic biology applications

Analytical method	Advantages	Disadvantages
FACS	Ultra-high throughput for fluorescence of single cells (up to 10^5 cells/s) Multi-parameter analysis of cell size, morphology, and fluorescence Rapid sorting	Inability to track temporal processes in single cells Limited to fluorescence-based intracellular product analysis
Beacon® Optofluidic system	Track and assay of the individual cells across multiple time points Automated and well-controlled culture of multiple single cells in compartments Capable of analyzing both intracellular and extracellular products	High cost Moderate throughput (lower than FACS, higher than microplate)
Droplet fluidics with cells	Reduced sample & reagent consumption Minimized Sample preparation (pre-culture) time Enabled cell-to-cell communication-based studies and screenings Capable of integrating label-free detection analysis (Raman, etc.) Capable of analyzing both intracellular and extracellular products	Required microfluidics expertise Moderate throughput
Droplet fluidics with CFPS	Applicable to non-model hosts Rapid testing w/o plasmid assembly, transformation, or genomic integration Serves as a model for studying gene expressions in defined environments Reduced sample & reagent consumption Capable of integrating label-free detection analysis (Raman, etc.)	Required microfluidics expertise Moderate throughput

FACS: fluorescence-activated cell sorting

utilizing calcein-filled sensor liposomes to intensify fluorescence upon permeabilization as non-ribosomal peptide gramicidin S production escalated. Table 2 summarizes examples of droplet microfluidics applications for single-cell analysis.

Moreover, this technology holds significance in cell-to-cell communication studies, enabling the correlation of extracellular metabolite concentrations with biosensor cell output through the coculture of producer and biosensor cells within the same droplets [51–55]. It has applications in screening enzymes from natural samples, where recombinant expression in traditional laboratory strains is often unsuccessful. An example of its applicability was demonstrated using GESS cells to detect soluble methane monooxygenase (sMMO) activity in methanotrophs, underscoring the potential of this approach for invaluable screening of naturally occurring microorganisms possessing sMMO activity [51]. Moreover, this technology holds potential for engineering microbial strains that fail to retain metabolites within cells. Siedler et al. [52] encapsulated yeast cells producing p-coumaric acid with *Escherichia coli* biosensor cells in droplets and successfully isolated target yeast cells producing

Table 2 Examples of droplet microfluidics applications for single-cell analysis

Target	Host	Library	Sensing	Sorting	References
α-amylase secretion	*Saccharomyces cerevisiae*	Random mutagenesis (UV)	Fluorogenic substrate	FADS	[42]
Cellulase	*Streptomyces lividans* 66	Random mutagenesis	Fluorogenic substrate	FADS	[44]
Glucose oxidase	*S. cerevisiae*	Saturation mutagenesis	ViPer assay	FADS	[45]
L-lactate production	Cyanobacterium *Synechocystis* sp. PCC6803	Random mutagenesis (UV)	Fluorescent dye	FADS	[46]
Megasynthetase (NRPS)	*Escherichia coli*	Site-specific mutagenesis	Calcein-filled liposome sensor	FADS	[43]
Nanobody VHH secretion	*Corynebacterium glutamicum*	CRISPRi library	FlAsH-tetracysteine assay	FADS	[47]
CAZyme	*E. coli*	Mock library	Coupling reaction mixture	AADS	[48]
Algal diacylglycerol acyltransferases	*S. cerevisiae*	Mock library	Label-free (Raman)	RADS	[49]
Phenylalanine ammonia lyase	*E. coli* Nissle	Site-specific mutagenesis	TF-based biosensor	FACS (W/O/W droplet)	[50]

NRPS nonribosomal peptide synthetases, *CAZyme* carbohydrate-active enzymes, *ViPer* vanadium bromoperoxidase-coupled fluorescence assay, *TF* transcription factor, *FADS* fluorescence-activated droplet sorting, *AADS* absorbance-activated droplet sorting, *RADS* Raman-activated droplet sorting, *FACS* fluorescence-activated cell sorting substantial amounts of extracellular p-coumaric acid using the fluorescent signal of *E. coli* [52]. Examples of droplet microfluidics applications for cell-to-cell communication based analysis were summarized in Table 3.

Despite its potential, the utilization of droplet microfluidics has not fully met expectations, mainly due to the challenges in precise and rapid sorting. Various research groups have worked consistently to overcome these hurdles. A particular challenge arises from the incompatibility of the conventional water-in-oil droplet format with FACS. An emerging solution involves the use of water–oil-water double emulsions that can be suited with FACS, potentially facilitating faster screening due to the inherent advantages of FACS [56–58]. However, achieving the necessary stability in these droplets, which is critical for efficient sorting and recovery, is practically challenging. It necessitates the delicate control over various forces—inertial, capillary, viscous, and shear—that interact at two separate interfaces during droplet formation, processing, and flow cytometry [59].

Another approach is the use of hydrogel droplets, where hydrogel reagent like agarose, collagen, gelatin, and poly(ethylene glycol) are incorporated during droplet synthesis [60]. After encapsulation, the hydrogel forms, and upon breaking the droplets, these hydrogel particles can be sorted using FACS. It is essential to ensure that the gelation reaction is suppressed during droplet synthesis, that stability is maintained during sorting, and that the gel can be easily degraded during the recovery process. Securing a reversible gel material is crucial in this context. For instance, low-melting agarose gel can undergo gelation at relatively low temperatures, allowing for encapsulation either at room temperature or slightly above (around 37℃) where cell damage is minimal [61]. However, external temperature variations, exposure durations can influence gelation. This sensitivity can result in clogging issues during droplet formation, potentially limiting the feasibility of prolonged, and robust screening.

Rather, the most widely used method is the highly efficient microfluidic fluorescence-activated droplet sorter (FADS). It is designed for emulsion droplets that can be sorted using DEP in a fluorescence-activated manner, similar to FACS. A notable example, the DEP sorter, is capable of analyzing and sorting droplets at a speed of 30 kHz and achieves over 99% accuracy [62]. Although it offers superior throughput, its application is limited owing to its small droplet size. One study overcame this limitation by developing a sequentially addressable dielectrophoretic array, enabling analysis and classification at 4.4 kHz speed with accuracy exceeding 98.8% for 57 µm droplets [63]. Nonetheless, efforts by various research groups to tackle these challenges and enhance the utility of droplet microfluidics through improved sorting techniques and enhanced stability of sample incubation within droplets continue, with its throughput (typically 10^{2-3}/s) yet to be further improved.

4 Droplet-based microfluidics and cell-free reaction

The integration of droplet-based microfluidics with CFPS systems offers a substantial opportunity to accelerate the DBTL process. Cell-free prototyping enables rapid testing without plasmid assembly, transformation, or genomic integration, making it particularly advantageous for investigating non-model or slow-growing organisms [64]. Native cell extract-based CFPS systems facilitate the characterization of proteins expressed in specific organisms, notably for the quick prototyping of new genetic regulatory parts in organisms such as *Bacillus subtilis* [65], clostridia [66], cyanobacteria [67], and mammalian cells. The efficiency and practicality of this approach can be enhanced using laboratory automation systems for small-volume samples or droplet microfluidic systems, enabling a lower sample volume and reagent costs, precise manipulation, and higher analysis throughput. Furthermore, this integration addresses the challenges related to substrate permeability and host compatibility in biocatalyst and/or cell factory development, ultimately

Table 3 Examples of droplet microfluidics applications for cell-to-cell communication based analysis

Target	Host	Library	Sensing	Sorting	Ref
Soluble methane monooxygenase (sMMO)	Methanotroph	Natural sample	*E. coli* with TF-based biosensor	FACS (agarose gel)	[51]
p-coumaric acid production	*Saccharomyces cerevisiae*	Mock library	*E. coli* with TF-based biosensor	FADS	[52]
3-hydroxypropionic acid (3-HP)	*Escherichia coli*	Randomly overexpressed endogenous genes	*E. coli* with TF-based biosensor	FADS	[53]
Erythromycin production	*Saccharopolyspora erythraea*	Random mutagenesis (ARTP)	*E. coli* with TF-based biosensor	FADS	[54, 55]

ARTP atmospheric and room temperature plasma, *TF* transcription factor, *FACS* fluorescence-activated cell sorting, *FADS* fluorescence-activated droplet sorting broadening the range of targets available for screening and exploration.

In particular, CFPS systems provide a streamlined strategy for accelerating the engineering of proteins and enzymes [68–73]. Gan et al. [68] demonstrated this using a high-throughput platform that incorporated DNA libraries, CFPS, droplet microfluidics, and next-generation sequencing, facilitating the rapid screening of genetic components to enhance regulatory elements in synthetic biology. In the field of enzyme engineering, Zhang et al. [69] achieved high-throughput protein synthesis and screening using femtoliter droplet arrays, resulting in a significant improvement (up to approximately 20-fold) in alkaline phosphatase activity. Another recent study reported a microfluidics-based in vitro evolution platform that leveraged picoliter-volume droplets to screen protease variants with higher activity using fluorogenic indicators [70]. These indicators fluoresce upon protease-mediated substrate cleavage. Using automated microfluidic sorting, the correlation between fluorescence and protease activity highlighted promising avenues for efficient genotype identification. Furthermore, a paramagnetic bead-based kinase screening platform has been developed that enables the identification of mutations in which leucine and isoleucine serve as anchors for facilitating mutations in non-preferred amino acids through positive epistasis [71]. Examples of droplet microfluidics applications for cell-free reaction based analysis were summarized in Table 4.

The integration of CFPS with microfluidic systems can serve as a model system and enable the quantitative investigation of the biochemical reactions of gene expression and protein synthesis dynamics within well-defined environments, unlike the complex cellular environment. For example, Kapsner and Simmel [74] studied transcriptional noise within oil-encapsulated microdroplets, which was significantly dependent on the template-to-polymerase ratio. In another study, transcription and translation dynamics were assessed in individual synthetic cells using fluorescent RNA aptamers and protein reporters in monodisperse liposomes incorporating a double emulsion [75]. This finding can expand the potential of engineered micron-sized compartments and integrated reaction networks guided by mathematical modeling, enabling intricate multiscale chemical system design from a fundamental approach. Furthermore, bacterial cells are not limited to model development, and a HeLa-based mammalian CFPS system was demonstrated using a locked nucleic acid probe for real-time transcription monitoring [76].

5 Label-free analysis

In addition to fluorescence-based techniques, label-free methods such as high-throughput mass spectrometry, Raman spectroscopy, and surface-enhanced raman scattering provide accurate methods for analyzing enzymes and cell factories. These methods enrich the analytical landscape, enabling the precise characterization and quantification of various molecules. Raman spectroscopy-based approaches, known for their accuracy and speed, have been effectively implemented [17, 18], such as in the screening of astaxanthin-enhancing cell factories [77]. However, integrating them with droplet microfluidic systems can be complex because of droplet curvature and background signals. Kim et al. [78] demonstrated optimized microscope optics that enabled the collection of Raman spectra from single cells within droplets in a microfluidic device. However, improvements are required to link these techniques effectively to droplet microfluidic systems for versatile applications in synthetic biology.

6 Future perspectives

Biofoundry leverages the capabilities of automated robotics to execute an efficient iterative DBTL cycle for the development of industrial biocatalysts and cell factories.

Table 4 Examples of droplet microfluidics applications for cell-free reaction based analysis

Target	Kit	Library	Sensing	Sorting	References
Alkaline phosphatase	PURExpress® (NEB)	Single-site saturation mutagenesis	Fluorogenic substrate	Manual sorting (microcapillary)	[59]
Protease	PURExpress® (NEB)	Random mutagenesis	Fluorogenic casein substrate	FADS	[60]
Kinase	PURExpress® (NEB)	Single-site saturation mutagenesis	Functionalized paramagnetic beads	FACS (bead)	[61]
Formate dehydrogenase	PURExpress® (NEB)	Site saturation	Functionalized paramagnetic beads	FACS (bead)	[77]
Aminoacyl-tRNA synthetase	The PURE system (in-house prepared)	Random mutagenesis	Amber stop codon-containing GFP mRNA	FACS	[78]

FADS fluorescence-activated droplet sorting, *FACS* fluorescence-activated cell sorting

By using CFPS tool in phenotyping and screening, the "Build" process can be accelerated by eliminating time-consuming procedures such as plasmid assembly, transformation, or genomic integration. Furthermore, CFPS stands out for its adaptability, facilitating research on non-model or slow-growing organisms. It also addresses notable challenges in cell-based screenings, like the engineering of enzymes for substrates inaccessible to cells, and producing proteins that may be inherently toxic. The integration of robotics, notably the Echo acoustic liquid handler, has enhanced reproducibility and precision in CFPS-based analyses, simultaneously minimizing human errors.

However, for the full realization of CFPS-based screening potential, integration with high-throughput testing and sorting equipment, such as FACS, is essential. The integration of microfluidics, specifically the hydrogel droplet technology, with FACS holds the potential to streamline the 'Test' process. Achieving this necessitates the refinement of practical and precisely controllable reversible gels to enhance testing efficiency. Another existing challenge is the requisite expertise in microfluidics. In this context, the approaches by DAMP Lab offer a valuable roadmap, suggesting ways to lower the barrier to droplet microfluidics utilization and setting the stage for its wider application across various research areas.

7 Conclusion

In conclusion, this comprehensive review underscores the pivotal role of microfluidic technologies in advancing high-throughput phenotyping and screening in the field of synthetic biology. While commercial microfluidic devices such as FACS and Beacon have accelerated single-cell analyses, they come with inherent limitations in terms of scalability and applicability. In contrast, droplet-based microfluidics, especially when integrated with CFPS, offers a flexible platform with a wide range of applications, bypassing traditional gene transfer constraints and ensuring precise micro-reaction environments. By comparing the strengths and weaknesses of commercial versus droplet-based microfluidic systems, researchers can harness them optimally, significantly expanding synthetic biology applications.

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Data availability Data will be made available on request.

Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical approval Neither ethical approval nor informed consent was required for this study.

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